Linear Simultaneous Equations

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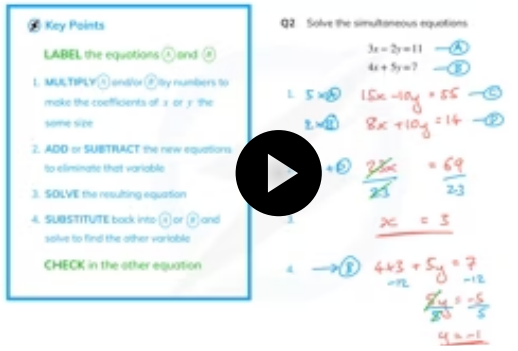

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Linear simultaneous equations

What are linear simultaneous equations?

- When there are **two unknowns** (***x* and *y***), we need **two equations** to find them both
  - For example, $3x + 2y = 11$ and $2x - y = 5$
    - The values that work are $x = 3$ and $y = 1$
- These are called **linear simultaneous equations**
  - **Linear** because there are no terms like x^2 or y^2

How do I solve linear simultaneous equations by elimination?

- **Elimination** removes one of the variables, x or y
- To eliminate the x's from $3x + 2y = 11$ and $2x - y = 5$, make the number in front of the x (the **coefficient**) in both equations the same (the sign may be different)
  - Multiply **every term** in the **first** equation **by 2**
    - $6x + 4y = 22$
  - Multiply **every term** in the **second** equation **by 3**
    - $6x - 3y = 15$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

eliminate is the **same**, **subtract** the equations

$$\begin{array}{r} 6x + 4y = 22 \\ -\ \underline{6x - 3y = 15} \\ 7y = 7 \end{array}$$

- The y terms have become $4y - (-3y) = 7y$ (be careful with **negatives**)
  - **Solve** the resulting equation to find y
  - $y = 1$

- Then **substitute** $y = 1$ into one of the **original** equations to find x
  - $3x + 2 = 11$, so $3x = 9$, giving $x = 3$

- Write out **both solutions** together, $x = 3$ and $y = 1$
- **Alternatively**, you could have eliminated the y's from $3x + 2y = 11$ and
  $2x - y = 5$ by making the **coefficient** of y in both equations the same
  - Multiply every term in the second equation by 2
  - **Adding** this to the first equation **eliminates *y*** (and so on)
    - When the **sign** in front of the term you want to eliminate is **different**, **add** the equations

$$\begin{array}{r} 3x + 2y = 11 \\ +\ \underline{4x - 2y = 10} \\ 7x \qquad = 21 \end{array}$$

How do I solve linear simultaneous equations by substitution?

- **Substitution** means substituting one equation into the other
  - This is an **alternative** method to **elimination**
    - You can still use elimination if you prefer

- To solve $3x + 2y = 11$ and $2x - y = 5$ by substitution
  - **Rearrange** one of the equations into ***y* =** ... (or x = ...)
    - For example, the second equation becomes $y = 2x - 5$

  - **Substitute** this into the first equation
    - This means **replace** all y's with $2x - 5$ in brackets
    - $3x + 2(2x - 5) = 11$

  - **Solve** this equation to find x
    - $x = 3$

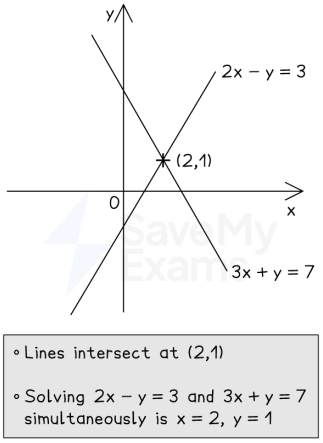
  - Then **substitute** $x = 3$ into $y = 2x - 5$ to find y
    - $y = 1$

How do I solve linear simultaneous equations graphically?

- **Plot** both equations on the same set of axes
  - To do this, you can use a **table of values**
    - or rearrange into $y = mx + c$ if that helps

- Find where the lines **intersect** (cross over)

- For example, to solve $2x - y = 3$ and $3x + y = 4$ simultaneously
  - First **plot** them both on the **same axes** (see graph)
  - Find the **point of intersection**, (2, 1)
  - The solution is $x = 2$ and $y = 1$



Examiner Tips and Tricks

Always check that your final solutions satisfy **both** original simultaneous equations!

Write out both solutions (x and y) together at the end to avoid examiners missing a solution in your working.

Worked Example

Solve the simultaneous equations

$$5x + 2y = 11$$
$$4x - 3y = 18$$

Answer:

It helps to number the equations

$$5x + 2y = 11 \quad (1)$$
$$4x - 3y = 18 \quad (2)$$

We will choose to eliminate the y terms
Make the y terms equal by multiplying all parts of equation 1 by 3 and all parts of equation 2 by 2

$$15x + 6y = 33 \quad (3)$$
$$8x - 6y = 36 \quad (4)$$

The $6y$ terms have different signs, so they can be eliminated by adding equation 4 to equation 3

Solve the equation to find x (divide both sides by 23)

$$x = \frac{69}{23} = 3$$

Substitute $x = 3$ into either of the two original equations

$$1 \quad 5(3) + 2y = 11$$

Solve this equation to find y

$$\begin{aligned} 15 + 2y &= 11 \\ 2y &= 11 - 15 \\ 2y &= -4 \\ y &= \frac{-4}{2} = -2 \end{aligned}$$

Substitute $x = 3$ and $y = -2$ into the other equation to check that they are correct

$$\begin{aligned} 2 \quad 4x - 3y &= 18 \\ 4(3) - 3(-2) &= 18 \\ 12 - (-6) &= 18 \\ 18 &= 18 \end{aligned}$$

Write out both solutions together

- This question can also be done by eliminating x first (multiplying equation 1 by 4 and equation 2 by 5 then subtracting)

$$x = 3, \ y = -2$$

How do I form simultaneous equations?

- Introduce **two letters**, x and y, to represent **two unknowns**
  - Make sure you know exactly what they stand for (and any units)
- Create **two different equations** from the words or contexts
  - 3 apples and 2 bananas cost £1.80, while 5 apples and 1 banana cost £2.30
    - $3x + 2y = 180$ and $5x + y = 230$
      x is the **price** of an apple, in **pence**
      y is the **price** of a banana, in **pence**
    - This question could also be done in pounds, £
- **Solve** the equations **simultaneously**
- Give answers **in context** (relate them to the story, with units)
  - $x = 40, y = 30$
  - In context: an apple costs 40p and a banana costs 30p
- Some questions don't ask you to solve simultaneously, but you still need to

- Solve simultaneously to get $x = 12$, $y = 7$
- The product is $xy = 12 \times 7 = 84$

Examiner Tips and Tricks

Always check that you've answered the question! Sometimes finding x and y isn't the end.

E.g. you may have to state a conclusion related to the context of the problem.

Worked Example

At a bakery a customer pays £9 in total for six bagels and twelve sausage rolls.

Another customer buys nine bagels and ten sausage rolls, which costs £12.30 in total.

Find the cost of 5 bagels and 15 sausage rolls.

Answer:

The two variables are the price of bagels, b, and the price of sausage rolls, s

Write an equation for the first customer's purchases, and label it equation 1

$$(1) \quad 6b + 12s = 9$$

Write an equation for the second customer's purchases, and label it equation 2

$$(2) \quad 9b + 10s = 12.3$$

We will choose to eliminate the b terms
Make the b terms equal by multiplying all parts of equation 1 by 3 and all parts of equation 2 by 2
Label these as equations 3 and 4

$$(1) \times 3 \quad 18b + 36s = 27 \qquad (3)$$
$$(2) \times 2 \quad 18b + 20s = 24.6 \qquad (4)$$

To eliminate b, subtract equation 4 from equation 3

$$(3) - (4) \quad 16s = 2.4$$

Solve for s

$$s = \frac{2.4}{16} = 0.15$$

Substitute this into either equation to find b, we will use equation 1

$$b = 1.2$$

So sausage rolls cost £0.15 each and bagels cost £1.20 each

Use these values to find the price of 5 bagels and 15 sausage rolls

$$(5 \times 1.2) + (15 \times 0.15) = 8.25$$

£8.25

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